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A case-specific psychological first aid AI recommendations app for rescue and first responders

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ABSTRACT

Background: Psychological First Aid (PFA) is an intervention designed to reduce the initial distress caused by traumatic events. The various types of PFA interventions have gained widespread acceptance in response to mass trauma and domestic violence; however, many first responders are often new to disaster scenes and have limited training and experience in providing PFA.

Objective: This article showcases the development of an AI-based chatbot that offers case-specific real-time recommendations in three languages to first responders for implementing PFA in accordance with established PFA models.

Method: Prioritizing ethical considerations and expert input, the PFA chatbot is designed for use by first responders, not victims, and offers case-specific guidance by matching user queries with a handcrafted dataset of trauma cases, ensuring alignment with accepted PFA intervention models. Semantic search techniques allow the chatbot to find the closest matching case and adapt the expert-provided response to the specific user query, leveraging few-shot learning to maintain relevance. Finally, a refined prompt engineering process ensures the chatbot's recommendations adhere to the accepted PFA protocols, with ongoing optimization to enhance response accuracy.

Results: Response times ranged from five to seven seconds, depending on Wi-Fi conditions. The chatbot provides structured, contextually appropriate PFA interventions, adapting to various traumatic scenarios for adults and children. It aligns with PFA protocols to deliver tailored recommendations that address the specific needs of each case.

Conclusion: The PFA chatbot enhances the accessibility and scalability of PFA intervention, offering standardized, case-specific interventions with minimal training. Its AI-driven system supports first responders in providing timely, high-quality care following traumatic events, making it particularly valuable in large-scale disasters. The tool improves outcomes by delivering tailored, practical recommendations based on established PFA protocols. Further research is required to determine the effectiveness of using the PFA chatbot in reducing ASR symptoms.

Aplicación de IA con recomendaciones de primeros auxilios psicológicos específicas para cada caso, dirigida a equipos de rescate y primera intervención

Antecedentes: Los primeros auxilios psicológicos (PAP) son una intervención diseñada para reducir el malestar inicial causado por eventos traumáticos. Los distintos tipos de intervenciones de PAP han alcanzado una amplia aceptación en respuesta a traumas masivos y violencia doméstica; sin embargo, muchos socorristas suelen ser novatos en situaciones de desastre y cuentan con limitada capacitación y experiencia en la prestación de PAP.

Objetivo: Este artículo muestra el desarrollo de un chatbot basado en IA que ofrece recomendaciones en tiempo real y en tres idiomas a los socorristas, específicas para cada caso, para la implementación de PAP de acuerdo con los modelos de PAP establecidos.

Método: Priorizando las consideraciones éticas y la opinión de expertos, el chatbot de PAP está diseñado para ser utilizado por personal de primera intervención, no por víctimas, y ofrece orientación específica para cada caso mediante la comparación de las consultas de los usuarios con un conjunto de datos de casos de trauma cuidadosamente elaborado, garantizando así su alineación con los modelos de intervención de PAP aceptados. Las técnicas de búsqueda semántica permiten al chatbot encontrar el caso más similar y adaptar la respuesta proporcionada por el experto a la consulta específica del usuario, aprovechando el aprendizaje con pocos ejemplos para mantener la relevancia. Finalmente, un proceso de ingeniería de mensajes refinado garantiza que las recomendaciones del chatbot se ajusten a los protocolos de PAP aceptados, con una optimización continua para

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PALABRAS CLAVE
Aplicación de primeros auxilios psicológicos con IA; psiquiatría avanzada; prevención de REA; primeros auxilios psicológicos

HIGHLIGHTS

- Many first responders lack sufficient training in providing Psychological First Aid (PFA) during disasters.
- An AI-based recommendation application was developed to deliver real-time, case-specific PFA recommendations in three languages, enhancing the capacity and confidence of first responders.
- The chatbot provides structured, tailored interventions with minimal training, improving outcomes in diverse trauma scenarios and large-scale disasters.

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mejorar la precisión de las respuestas.
Resultados: Los tiempos de respuesta oscilaron entre cinco y siete segundos, según las condiciones de la conexión Wi-Fi. El chatbot proporciona intervenciones de PAP estructuradas y adaptadas al contexto, ajustándose a diversos escenarios traumáticos tanto para adultos como para niños. Se alinea con los protocolos de PAP para ofrecer recomendaciones personalizadas que abordan las necesidades específicas de cada caso.
Conclusión: El chatbot de PAP mejora la accesibilidad y la escalabilidad de la intervención, ofreciendo intervenciones estandarizadas y específicas para cada caso con una capacitación mínima. Su sistema basado en IA apoya a los socorristas para brindar atención oportuna y de alta calidad tras eventos traumáticos, lo que lo hace especialmente valioso en desastres a gran escala. La herramienta mejora los resultados al proporcionar recomendaciones prácticas y personalizadas basadas en protocolos de PAP establecidos. Se requiere más investigación para determinar la eficacia del chatbot de PAP en la reducción de los síntomas de Reacción al Estrés Agudo.

The current article aims to present an AI-based recommendation chatbot to support the implementation of scientifically based psychological first aid (PFA). Traumatic events often lead to long-term consequences that impact personal, familial, and social functioning. The type and timing of the initial response to a traumatic event are critical for fostering future adaptation and resilience (Hobfoll et al., 2007; Wright et al., 2012). Psychological First Aid (PFA) is an intervention designed to reduce the initial distress caused by traumatic events, such as acute stress response (ASR, DSM-5 308.3; ICD-10, F43; ICD-11, QE84), aiming to buffer against and alleviate the long-term psychological and somatic effects of such events, including post-traumatic stress disorder (PTSD) (Everly & Lating, 2021; Forbes et al., 2010; Ruzek et al., 2007; WHO, 2011).

1. The rationale for AI-based PFA system development

Recent reviews highlight the expanding role of LLM-powered chatbots in medicine, with promising applications in medical writing, clinical documentation, education, and basic diagnostic support (Chow et al., 2023; Chow & Li, 2024). Models such as GPT-5 have shown particular potential in delivering empathetic, personalized, and context-aware assistance to healthcare providers (Chow & Li, 2024). Nonetheless, significant concerns remain regarding misinformation, outdated or biased training data, and the absence of medical-specific fine-tuning. Ethical challenges, including patient privacy, transparency, and equity, remain unresolved (Chow et al., 2023; Chow & Li, 2024). Additionally, beyond weak clinical validation, emerging guidance warns of high-consequence failure modes: automation bias and over-reliance can delay time-critical care, generate unsafe triage advice, and elicit harmful responses in psychiatric crises that may exacerbate self-harm risk (Abrams, 2025; Kämmer et al., 2024; WHO, 2021, 2025).

This article presents the development of the AI-PFA recommendation app, utilizing an expert-guided decision-support framework grounded in forward psychiatry (Jones & Wessely, 2003) and built with ethical AI principles (Chow et al., 2023; Chow & Li, 2024).

2. PFA interventions

Immediate provision of PFA following trauma is critical for addressing ASR, mitigating the risk of PTSD, and fostering survivors' resilience (Hobfoll et al., 2007; Solomon et al., 2005; Wright et al., 2012). Many first responders, including social workers and healthcare professionals, lack PFA training (Fox et al., 2012; Geoffrion et al., 2023; Prykhodko et al., 2021; Ruzek et al., 2007; Shultz & Forbes, 2013) and often rely on simplistic or emotionally-driven responses that may worsen the survivor's distress (Brymer et al., 2006). Additionally, vague and overly broad protocols make consistent implementation and adherence challenging (J. Bisson & Lewis, 2009; Wang et al., 2021). Moreover, the unpredictable nature of disasters and emergencies can overwhelm first responders, increasing their need for real-time expert guidance, as even the most well-trained staff cannot anticipate or be fully prepared for every possible situation (Forbes et al., 2011).

Additionally, a systematic review examining the programmatic effectiveness of PFA identified only 12 eligible studies from over 9000 reviewed citations. While preliminary findings suggest that PFA may contribute to improved mental health and psychosocial well-being among individuals exposed to potentially traumatic events, the review underscored major limitations in existing literature. These included inconsistent intervention components, insufficient evaluation methodologies, and high risks of bias, factors that significantly undermine the ability to draw robust conclusions about PFA's overall impact (Hermosilla et al., 2023). This article presents an AI-based tool

designed to guide first responders in applying empirically supported PFA techniques while also enabling the systematic assessment of PFA effectiveness.

PFA plays a dual role in supporting the immediate needs of trauma survivors and managing the stress of first responders. It is recognized as an emergency intervention that offers essential psychological support, which is critical for helping victims begin the recovery process in the aftermath of a traumatic event (Cain et al., 2010; Everly & Lating, 2021; Mendez et al., 2023; Wright et al., 2012). For trauma survivors, PFA can provide immediate relief, significantly reducing acute stress responses and offering support in the wake of traumatic events (Everly & Lating, 2021; Fox et al., 2012; Shultz & Forbes, 2013; WHO, 2011). Consequently, it is considered to have a significant influence on the victim's long-term adaptation and psychological health, particularly for children (Gilbert et al., 2021; Jayuphan et al., 2020). Research shows that when responders are prepared and guided, they can offer more effective care to trauma survivors, thereby reducing distress and preventing further trauma (Bisson, 2014; Dieltjens et al., 2014). In contrast, first responders under significant stress may react in ways that inadvertently traumatize survivors further(Greinacher et al., 2022) Studies indicate that providing structured PFA interventions can reduce ASR and improve psychological resilience among responders themselves (Bekircan et al., 2024; Wang et al., 2021), including in real-world disaster scenarios, such as volunteers responding to an earthquake disaster in Turkey (Bekircan et al., 2024).

3. PIES forward psychiatry framework

Scientific evidence and the widespread implementation of the PIES framework across organizations and countries led us to adopt it as the clinical foundation for the AI PFA system's API. Compared to other PFA interventions, the PIES model is supported by scientific evidence demonstrating its positive impact on ASR victims (Jones et al., 2007; Solomon et al., 2005; Solomon & Benbenishty, 1986). However, scholars have criticized the evidence backing PFA models as primarily anecdotal or derived from smaller studies lacking rigorous methodological standards (Hermosilla et al., 2023). Consequently, while the literature reflects a broad consensus among practitioners regarding its potential benefits, it does not provide substantial scientific support (Fox et al., 2012). Specific evaluations have indicated high risks of bias, particularly in studies examining the various components of PFA and their individual impacts on outcomes. (Wang et al., 2022). In contrast to this literature, the PIES 'Forward Psychiatry,' (Jones & Wessely, 2003) devised during World War I for the treatment of shell shock, remains the standard intervention.

The reintroduction of the PIE methods by US and UK forces during World War II, along with their successful application by Israeli forces in the Lebanon conflict, demonstrated significant efficacy in returning servicemen to active duty, with evaluations revealing higher return-to-duty rates than those at base hospitals. This suggests promising effectiveness in treating ASR and minimizing psychiatric casualties (Jones et al., 2007; Solomon et al., 2005; Solomon & Benbenishty, 1986). Later research conducted by Solomon et al. (2005) and Solomon and Benbenishty (1986) strongly demonstrated the efficacy of the PIES protocol in frontline treatment for combat stress reactions. Solomon and Benbenishty (Solomon et al., 2005; Solomon & Benbenishty, 1986) identified that the PIE approach played key roles in facilitating rapid recovery for soldiers experiencing acute combat stress reactions, highlighting how timely and localized interventions enhance adaptive coping and reduce the risk of chronic psychological issues. The study included several hundred Israeli soldiers diagnosed with combat stress reactions during the 1982 Lebanon War, representing 82% of the total target population, though exact numbers could not be disclosed due to security restrictions. The 20-year longitudinal evaluation by Solomon and colleagues (Solomon et al., 2005; Solomon & Benbenishty, 1986) (Total $N = 429$ participants at 20-year follow-up, including 79 in the frontline treatment group, 156 in the rear echelon treatment group, and 194 in the comparison group) further underscores the long-term benefits of the PIES principles, showing that early intervention grounded in the PIES framework results in better psychological outcomes and a lower incidence of PTSD. Specifically, 20 years after the war, soldiers who received frontline treatment according to PIE principles exhibited significantly lower levels of post-traumatic and psychiatric symptoms, reported less loneliness, and demonstrated better social functioning compared to similarly traumatized soldiers who did not receive such treatment. Furthermore, a cumulative effect was observed: the greater the number of frontline treatment principles applied, the more pronounced the positive impact on psychiatric outcomes. Although the PIES principles are widely endorsed, codified in NATO AMedP-8.6 (2019) and supported by operational experience (Solomon et al., 2005), the empirical base for PFA remains limited and heterogeneous. Existing studies are methodologically varied with mixed effects on mental health outcomes, underscoring the need for rigorous controlled evaluations to clarify effectiveness, mechanisms, and implementation conditions (Hermosilla et al., 2023).

Today, these principles are recognized and applied in events involving acute stress (Everly & Lasting, 2019). The PIES principles have been translated into PFA protocols by the WHO (2011) and the 'Six Cs'

(Farchi & Gidrom, 2023). The various types of PFA interventions have gained widespread acceptance in response to mass trauma and domestic events, due to their simplicity and applicability, without the need for specialized mental health training (WHO, 2011). However, as many first responders, including volunteers, police officers, social workers, and other emergency response personnel, are often new to disaster scenes (Bekircan et al., 2024; Scott & Coleman, 2016) and have limited training and experience in providing PFA (Motola et al., 2015; Pedersen et al., 2016). To enhance their ability to deliver PFA effectively in real-time situations, there is a need for an easy-to-use tool that can increase their professional capacity, confidence, and the impact of PFA on victims. This article describes the development of an AI-based recommendation application that offers case-specific real-time recommendations in three languages to first responders for implementing PFA in accordance with established PFA models.

The AI-based PFA recommendation application provides case-specific recommendations based on the accepted PFA models: the World Health Organization (WHO) and the 'SIX Cs' PFA intervention models. The WHO model (2011) focuses on calming individuals, helping them address basic needs, and connecting them to resources and information. In contrast, the 'SIX Cs' model (Farchi et al., 2018) emphasizes cognitive and behavioural protocols grounded in the neuropsychology of resilience, aiming to quickly shift individuals from a state of helplessness to active and effective coping. This model targets the reduction of acute stress responses through structured cognitive and behavioural interventions that enhance resilience and self-efficacy (Farchi et al., 2018; Levy et al., 2020).

Unlike the questionable effectiveness of psychological debriefing techniques (Vignaud et al., 2022), the SIX Cs model has demonstrated efficacy in specific settings, such as emergency rooms and among migrant populations, where rapid cognitive activation has been shown to significantly reduce acute stress reactions (Farchi & Gidrom, 2023). Cognitive activation refers to the intentional stimulation of the prefrontal cortex through structured decision-making or action-based choices during acute stress, which helps reduce perceived helplessness and restore a sense of control (Farchi et al., 2018). This engagement of the prefrontal cortex facilitates top-down inhibition of the amygdala, thereby modulating the neurobiological stress response and promoting functional coping (Amat et al., 2008; Taylor et al., 2008). Another feature of cognitive activation is framing the event for the survivor, which involves shaping the traumatic incident into a coherent and manageable narrative to promote adaptive functioning (Farchi et al., 2024; Levy et al., 2020). Providing clear and contextual information helps shift individuals from confusion and helplessness to active decision-making, a central mechanism of the Six Cs model. Effective framing offers psychological scaffolding that restores meaning, enhances a sense of agency, and supports a return to functional coping.

We clarify that PIES (Jones et al., 2007), SIX Cs (Farchi et al., 2018), and WHO PFA (2011) are distinct and not interchangeable. The WHO and SIX Cs PFA models can be viewed as complementary. The WHO model is generic and universally applicable, emphasizing immediate practical and emotional support, while the SIX Cs model provides a structured cognitive activation intervention. Consequently, the WHO model allows a broader range of individuals, including non-professionals, to administer PFA, whereas the SIX Cs model typically requires training. Additionally, the WHO model guides first responders in directing victims to resources and advising on actions to take in the days following a traumatic event. It also addresses basic needs such as food and sleep, which are crucial for all victims, especially children. By contrast, PIES sets operational principles for delivery in the field (proximity, immediacy, expectancy, simplicity) and guides staging, triage, and brief on-site stabilization rather than specifying the content of supportive conversations. Therefore, the PFA bot recommendations layer these models without conflation: WHO actions are the universal baseline; SIX Cs modules are available when indicated and when trained providers opt in; and PIES informs contextual rules about where, when, and how brief interventions are delivered. This integration enables first responders to provide support that addresses immediate and longer-term needs while preserving each model's distinct purpose. The PFA bot can offer evidence-based support, potentially reaching a wider audience with less dependency on ongoing training (Jung & Han, 2021; Wang et al., 2021), thereby increasing the quality of PFA implementation (Bakker et al., 2020; Roncero et al., 2020).

When delivered appropriately, PFA has been shown to effectively prevent or reduce ASR (Solomon et al., 2005; Solomon & Benbenishty, 1986), not only in the immediate minutes following a traumatic event but also when provided up to 48 h later (Farchi et al., 2018). The chatbot is therefore designed to be used during the critical first 48 h after exposure to trauma, a period in which the initial adrenaline response typically diminishes and symptoms of psychological shock are most likely to emerge (Mortimer & Mortimer, 2023; Takayanagi & Onaka, 2021). This approach aligns with the diagnostic criteria for ASR as defined in the DSM-5 (308.3) and ICD-10 (F43). Clinicians have reported that applying PFA *after* the adrenaline rush rather than during the traumatic event is crucial due to the physiological and

psychological states induced by the fight-or-flight response (Reeves, 2020). As the adrenaline wears off, the body transitions to a calmer state through the parasympathetic nervous system, making individuals more receptive to support and intervention. This period is critical for providing PFA, as the individual becomes more aware of their physical and emotional state, aligning with the body's natural recovery processes. Therefore, using the bot after the adrenaline rush, either on the way to the scene or within 48 h after the event, is suggested to decrease the likelihood or severity of ASR symptoms.

3.1. The development of the AI-powered PFA system

Three key ethical principles guided our methodology for developing the chatbot. First, the chatbot app is designed for use by first responders rather than individuals in distress, ensuring responsible AI application. Second, it was developed by nine experts who trained the prompts, ensuring that the recommendations were grounded in established PFA intervention models. During the proof of concept (POC), the team of professionals also generated and validated the case bank. Using a shared WhatsApp group for training, this team systematically validated and iteratively adjusted the chatbot's guidance to maintain fidelity to protocols aimed at reducing acute shock symptoms. The experts and professionals who helped train the bot were (1) mental health officers and senior commanders in the defence forces as the PIES protocol was originally developed within the framework of forward psychiatry; (2) scientists specializing in emergency medicine; (3) social workers and volunteers with expertise in emergency care affiliated with national emergency medical, disaster, ambulance, and blood bank service, (Knobler et al., 2020) (4) two certified PIES instructors affiliated with a university; and (5) first aid responders serving in frontline units of hospitals and NGOs. Finally, the chatbot functions as a decision support system, offering suggestions rather than replacing human responders. This approach enhances the ability of first responders to implement clinically-based PFA interventions effectively while maintaining the essential human element in crisis response.

During the POC the AI-powered PFA system was delivered through WhatsApp, which was initially chosen as the platform for the POC due to its accessibility for the first responders' context. This system integrated the API with a structured case bank and a rule-based reasoning engine to deliver real-time, protocol-guided PFA recommendations. The combination allows for dynamic language generation alongside consistent adherence to established PFA procedures. By merging large language model (LLM) capabilities with expert-verified case logic, the system functions as an ethical, scalable, and context-sensitive decision-support tool.

To develop the PFA case-specific recommendation chatbot, we employed the following steps to create an LLM that utilizes a handcrafted case description dataset along with advanced natural language processing (NLP) techniques to provide case-specific professional recommendations. Here is a detailed breakdown of the method:

1. User Query Initiation:
   - Input: A first responder end-user sends a query to the app.
   - Process: This query typically describes a traumatic situation that the first responder is currently handling. Additionally, the query is being pre-processed to fetch the user first name and gender if those exist.
2. Semantic Case Matching:
   - Input: The user's query.
   - Process: The user's query is sent to the LLM via an API to perform a semantic search across the cases dataset, identifying the most semantically similar previously handled case. The cases in the dataset consist of pre-defined queries and responses formatted according to the accepted PFA interventions. This step involves computing semantic distances between the new query and existing cases, focusing on finding the closest match based on contextual and factual content.
3. Response Formulation:
   - Input: The closest matching case and its associated expert response, along with the user's original query.
   - Process: Both the selected case and the new query are sent to the LLM via an API. The LLM then generates a tailored response that adapts the pre-existing expert response as a form of 'few-shot learning' to the specifics of the new situation (new query). This step leverages the model's capacity to integrate and reformulate information, ensuring that the advice is both relevant and specific to the new case while maintaining the original answer format.
4. Delivery of Recommendation:
   - Output: A recommendation that includes SIX Cs steps for effectively coping with the situation.
   - Process: The generated recommendation is formatted into a structured response and sent back to the conversation, allowing the first responder to immediately apply the guidance in the field.

This methodology ensures that the chatbot provides timely, relevant, accurate, and actionable guidance aligned with established PFA protocols, enhancing the first responder's ability to manage trauma effectively.

By integrating advanced NLP techniques with a carefully curated case dataset, the chatbot serves as a robust tool in the arsenal of emergency response strategies, facilitating better immediate care and potentially improving long-term outcomes for trauma victims.

4. Prompt engineering

Here is the prompt we used. The core concept of our method relies on prompt engineering to guide the chatbot in generating precise and contextually appropriate responses. To implement this, we carefully crafted prompts instructing the chatbot to respond based on specific user inputs and scenarios. For instance, the prompt we used incorporates several parameters to effectively tailor the chatbot's responses.

- Username: The name of the person making the inquiry is integrated multiple times within the response to personalize the interaction and maintain engagement.
- User gender: The gender of the person making the inquiry to ensure the response is appropriate to the user's gender.
- Example Case: The case retrieved from the case database serves as a model for the response to be generated.
- User Query: The initial query submitted by the user prior to any processing.

'You are a responder from rescue forces assisting people in distress. Your response to the caller must help those in distress return to routine and function with practical and detailed suggestions that aid them in effective coping. The goal is to help the person about the call concerns overcome shock or distress through action-oriented focus. Avoid giving psychological advice, discussing feelings, expressing empathy, identifying with the distressing situation, explaining that it is natural to feel this way, and using the words "trauma" or "traumatic".' +\

'\n Incorporate the caller's name several times in the text, the caller's name is: ' + user_name +

'\n Incorporate the caller's gender correctly in the text, the caller's gender is: ' + user_gender +

'\n Respond exactly as the person who answered in the following format according to the query, do not be creative: ' +\ example_case +

'\n This is the query: \n' +\ user_query

It is important to note that instructing first responders to avoid emphasizing emotional processing or normalizing the victim's distress is based on the principle of cognitive activation, a core mechanism of the 6Cs PFA model. Cognitive activation seeks to engage the prefrontal cortex through structured, task-oriented decision-making, thereby enhancing perceived control and reducing helplessness in the immediate aftermath of trauma. For this reason, the LLM used in the chatbot was configured to limit highly empathetic responses and instead provide guidance strictly aligned with the evidence-based principles of the PIES protocol.

Fine-tuning our methodology is a critical process in which we iteratively adjust the chatbot's prompts to ensure they closely align with the specific needs and nuances of psychological first aid. This refinement ensures that the chatbot's responses are not only precise but also highly relevant to the scenarios encountered by first responders. The optimization process involves using a test set and having a professional review the results after each iteration. By continually optimizing these prompts, we enhance the chatbot's ability to deliver targeted, effective guidance in real-time situations while maintaining strict adherence to the structured guidelines necessary for handling traumatic events. This ongoing adjustment process is vital for ensuring that the chatbot functions as an effective support tool under varying real-world conditions.

4.1. Case description database

Our method is anchored in a meticulously curated dataset of 20 real-world generic scenarios developed by professionals to refine and validate AI-driven recommendations in the medical field. Each case in the dataset has been carefully documented and addressed by experts specializing in trauma and emergency response. The data can be viewed using this hyperlink. The scenarios encompass a range of traumatic events, including victims exposed to acts of violence such as murder, casualties from car accidents, and individuals evacuated from catastrophic incidents like building collapses following earthquakes. For each scenario, experts have crafted tailored responses that align with the structured steps of the World Health Organization (WHO) and the SIX Cs of Psychological First Aid (PFA) models, ensuring that each response offers a practical, actionable strategy consisting of five key recommendations. This approach not only leverages real-case scenarios to enhance the AI's learning but also ensures that the responses adhere to proven intervention strategies, formats, and language, providing a solid foundation for the application of AI in medical recommendations.

In terms of design and development, our methodology utilises a Retrieval-Augmented Generation (RAG) approach that draws on external data sources as inputs to the LLM. In RAG, a dedicated knowledge base is queried for relevant material; the retrieved content is embedded as high-dimensional vectors and incorporated into the model's generative process. While this yields precise, factual guidance, it also means output quality depends heavily on retrieval and may constrain generative flexibility. To balance

representativeness with ethical control, we curated a non-duplicative library of 20 prototypical emergency ASR scenarios. Limiting the corpus to these representative cases enhances alignment with the protocol, enhances case matching, reduces spurious retrieval, and improves cross-site reproducibility. Our decision to confine the library to 20 scenarios aligns with design principles seen in recent mental-health RAG evaluations that employ modest, bounded sample sets or guideline corpora (Amugongo et al., 2025; Bilal et al., 2025; Cremaschi et al., 2025; Jin et al., 2025; Masanneck et al., 2025; Ravenda et al., 2025).

Our method employs a handcrafted case description dataset not merely for information retrieval but as a contextual template for response generation. When a new trauma case is presented, our bot uses semantic search to find the closest matching case in the example dataset. Instead of converting this information into vectors, the entire matched case, including its detailed response, is utilized. This retrieved response serves as a guideline, enabling the AI to formulate a new, tailored response that adheres to the principles and structured steps of the SIX Cs model.

This method allows for greater flexibility and precision in generating responses, as it can adapt the principles of the example response to fit the unique aspects of the new case. By using our handcrafted case description dataset as a contextual template rather than a direct source of information, our approach ensures that responses are not only contextually relevant but also meticulously aligned with strategies for addressing trauma and anxiety. This approach prioritizes adherence to the structured format of the SIX Cs, ensuring consistency and efficacy in the mental aid provided, unlike RAG, which is limited by its reliance on direct factual retrieval and integration.

The AI-powered PFA system is embedded is a messaging-based Software-as-a-Service (SaaS) platform, accessed via a dedicated easy-to-use app. As a web-based application, it requires no installation; users access it via a browser and may optionally add a shortcut to their smartphone or computer home screen. Importantly, the system has been tested successfully under both Wi-Fi and cellular network conditions, and it does not require a Wi-Fi connection to operate, enhancing its utility in field environments where connectivity may be limited. The backend is hosted on Amazon Web Services (AWS) in Frankfurt, within a Virtual Private Cloud (VPC) infrastructure. It employs SSL encryption, dual-factor authentication, and load balancing to ensure security, performance, and scalability. All demonstration data were encrypted and managed in compliance with medical data standards, with only the first responder's phone number retained for minimal identification.

The AI PFA application is a decision-support tool for first responders, not for survivors, and is intended to complement, rather than replace, professional judgement and established protocols, in line with WHO guidance that PFA constitutes an initial response which may require subsequent referral (WHO, 2011). It is therefore not a self-help tool. As such, first responders should be appropriately prepared and operate within role boundaries (WHO, 2011). Where connectivity or access problems would delay the response or divert attention, standard operating procedures should proceed without the tool (IASC, 2007). Risk mitigation is embedded insofar as the recommendations align with PFA principles, reflected in historical and contemporary guidance of onward linkage to services (Jones et al., 2007; Jones & Wessely, 2003; NATO, 2019; WHO, 2011). The application should not be used when the scene is unsafe or there is an ongoing threat, as life-safety and command procedures take precedence and digital tools must not delay evacuation or hazard control (WHO, 2011). Nor should it be used during time-critical medical emergencies, where immediate triage or resuscitation must not be impeded by psychosocial tasks (IASC, 2007; WHO, 2011). In circumstances of severe psychiatric or behavioural risk requiring urgent care, such as violent agitation, suicidal attempt, or medical instability, local emergency pathways and specialist care should be activated before any consideration of using the app (Jones & Wessely, 2003; NATO, 2019).

5. Findings

5.1. Evaluation of generated responses

The primary challenge in prompt engineering was refining the large language model to deliver recommendations strictly aligned with the accepted PFA models: the 6Cs and the WHO. The process of engineering and training the prompts spanned eight weeks, during which we incorporated over 80 real-world cases submitted by first responders into the bot. Each set of recommendations was carefully reviewed, and the prompt instructions were adjusted to ensure precise responses. The final answers were validated by nine PFA experts in social work and emergency medicine, ensuring the model's accuracy and reliability. We optimized response time by streamlining the case dataset, reducing it to 20 generic cases without compromising the accuracy or relevance of the recommendations. After integrating the chatbot into WhatsApp for the POC, response time varied from five to seven seconds.

5.2. Examples of generated responses

Here are three different cases of the inquiry, along with their recommendations. Please refer to the Appendix to review the full response and

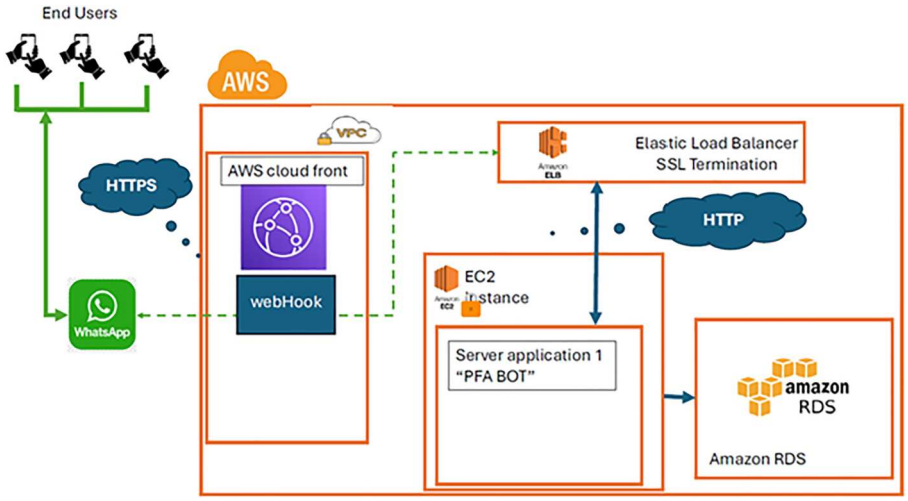


Figure 1. AI-based PFA bot architecture.

recommendations. We entered the WhatsApp test number for real new events provided by social workers and paramedics from the rescue forces.

1. Adult experiences the traumatic event – earthquake[1]
   - Case Description: 'I'm arriving after two buildings collapsed after an earthquake, finding a woman uninjured but in shock and unable to speak.'
   - Generated Response: PFA chatbot provides a step-by-step guide for Shalom to assist the woman, including (a) framing the event ('Explain to her that the event has ended'), (b) making eye contact and ask for her cooperation, (c) asking specific questions, (d) assigning tasks, (e) suggesting aftercare ('Explain to her that she needs to stay active and report who will be with her in the coming days and who she can call if she has difficulty performing tasks'), and (f) offering encouragement.
2. A child witnessing/exposed to a traumatic death of a parent[2]
   - Case Description: Talia describes the child witnessing his mother being stabbed by his father; he is in complete shock and shivering.
   - Generated Response: PFA chatbot advises Talia to (a) frame the event for the child,[3] (b) make eye contact, reassure her intention to help and ask for the child's cooperation, (c) ask thought-provoking questions, (d) assign simple tasks, (e) maintain activity and find support (e.g. 'find out who will be with him in the upcoming days'), and (f) provide encouragement.
3. Children's secondary exposure to death[4]
   - Case Description: Danny arrives at a scene and describes two children who witnessed a man falling from a crane near their school. The children are in shock and unable to speak.
   - Generated Response: PFA chatbot instructs Danny: to (a) frame the event, (b) make eye contact, reassure the children, and ask for cooperation, (c) ask engaging questions, (d) assign simple tasks, and (e) coordinate further support with the children's parents ('report to the children's parents and decide together on further support').

6. System overview and architecture

The PFA Bot system is built as a secure, cloud-based service that allows first responders to receive real-time, case-specific guidance via an easy-to-use messaging interface. In our current implementation, the chatbot is accessed through an easy-to-use three language web app, but the system is flexible and can be connected to other platforms. Figure 1 shows how the system works: when a user sends a message (such as a description of a victim in an emergency event), it is securely transmitted via HTTPS to a central cloud service hosted on Amazon Web Services (AWS). This message passes through an Elastic Load Balancer, which helps manage traffic and ensures the system runs smoothly even with many users. From there, it reaches a secure virtual server (EC2 instance), where the core chatbot application operates. This application processes the user's message using advanced AI techniques and retrieves the most relevant recommendation, which is then sent back to the user. All messages and responses are stored securely in a relational database (Amazon RDS), allowing for later analysis and improvement of the system.

7. Discussion

This article aimed to showcase the development of the first AI-powered PFA chatbot designed to provide PFA recommendations for first responders. The

findings demonstrate the chatbot's ability to deliver structured, consistent, and contextually appropriate PFA interventions based on the WHO and SIX Cs models. Additionally, the chatbot provides tailored psychological first-aid recommendations for specific cases. It has shown adaptability across various traumatic scenarios, effectively delivering standardized PFA protocols that address the unique needs of both adults and children. The integration of established PFA protocols allows the chatbot to effectively generate practical, context-sensitive recommendations.

Our discussion of the performance of the PFA chatbot and its potential significance should be viewed within the following limitations. First, our PFA Bot suffers from inherent biases in the dataset and training. If the data contains biases, these can influence the recommendations it provides. Since the bot is not a diagnostic tool but rather gives recommendations based on similar past situations, it can avoid some of the biases that affect diagnostic AI tools. However, it still relies on human responses from its training data, which can carry their own biases based on assumptions or worldviews. To ensure fair and accurate recommendations, it is crucial to continually monitor and update the training data to minimize these biases. Expanding the dataset to include a more diverse range of trauma scenarios and contexts will further improve the bot's performance.

Second, the chatbot struggles to provide clear recommendations for simple tasks, often skipping directly to more advanced steps, such as seeking external help. Additionally, it may offer inappropriate or problematic suggestions when adhering to the restrictions established in prompt engineering and following the closest case from the dataset. For example, in a situation where a child witnessed a parent's death, the bot might suggest asking, 'How long did you know your parent?' – a question that could be emotionally harmful. To address this, we are considering implementing an adaptive content filter for specific cases involving exposure to violence towards family members and increasing the number of cases in the dataset to better accommodate nuanced situations and tailored scenario-specific responses to prevent inappropriate recommendations.

Digital health tools, including e-health applications and AI-powered mental health support systems, have shown promise in enhancing the quality and accessibility of healthcare (Bakker et al., 2020; Roncero et al., 2020). However, existing tools and research predominantly focus on either emergency healthcare solutions, which do not address psychological needs, or mental health tools designed for after-care support of trauma survivors, which fail to provide solutions for emergency PFA scenarios. Our PFA initiative addresses this gap by delivering real-time responses for first responders, including both volunteers and trained mental health and rescue personnel. The PFA chatbot offers the advantage of accessibility with minimal training, while its AI-based decision-making support system ensures high-quality standardized PFA guidelines.

A wide range of non-emergency e-health applications have been implemented and have proven effective for accessible and efficient interventions, including those aimed at trauma survivors (Asselbergs et al., 2018; Bakker et al., 2020; Carr et al., 2018). The PFA completes the care continuum by providing protocol-based, case-specific guidance following exposure to traumatic events, thereby enhancing the professional capacity of first responders to provide timely PFA interventions based on the bot's recommendations.

Moreover, AI-powered tools, such as chatbots, have demonstrated potential in enhancing the scalability and personalization of mental health support. The PFA Chabot, as along with other AI-powered mental health tools, can facilitate real-time data analysis and feedback, allowing for tailored interventions that meet individual needs and improve overall outcomes (Cernvall et al., 2018; Torous et al., 2021). While the potential for AI tools in emergency healthcare is significant, only a few have been implemented, largely due to challenges such as potential bias in datasets (Mueller et al., 2022; Roncero et al., 2020). Overall, these advancements have the potential to make PFA more efficient, accessible, and effective, ensuring timely support for both survivors and responders in various contexts.

Our PFA chatbot has the potential to impact the field of psychotraumatology significantly. Traditional PFA relies heavily on the availability and readiness of trained responders, who are crucial in offering immediate relief and supporting long-term recovery for trauma survivors (Everly & Lating, 2021; Fox et al., 2012; Shultz & Forbes, 2013; WHO, 2011). However, their effectiveness is often constrained by the need for timely guidance and ongoing training and by the unpredictability of emergency situations (Forbes et al., 2011; Shultz & Forbes, 2013). In contrast, our chatbot provides automated, consistent, and protocol-based support, reducing dependency on human resources and ensuring immediate availability during large-scale disasters or in areas with limited access to mental health professionals. This scalability addresses significant gaps identified in the literature regarding the timely and effective delivery of PFA interventions (Fox et al., 2012; Geoffrion et al., 2023; Prykhodko et al., 2021; Ruzek et al., 2007; Shultz & Forbes, 2013).

As such, the bot could also be used as part of first responders' training. Utilizing the bot enhances the accessibility and scalability of PFA, particularly in large-scale disasters where human resources are limited. Practically, the tool offers standardized PFA recommendations for first responders that could be implemented across various fields while requiring

minimal training. This has significant benefits for trauma survivors, as it ensures timely and appropriate interventions. The accessibility of the tool allows more people to receive high-quality PFA. As shown in the examples, the tool provides simple, clear suggestions that require minimal training, making it accessible to a wide range of users, from professionals to volunteers.

Additionally, future versions of the bot could incorporate follow-up suggestions for responders, refer them to further support, and ask about the effectiveness of the intervention. This adaptive capability is a significant advantage over static training programmes. This approach enhances the practicality and immediacy of PFA delivery, addressing a critical need identified in previous studies (Jung & Han, 2021; Wang et al., 2021).

8. Future research directions

To validate real-world efficacy and address potential limitations, future work will evaluate the bot's implementation and impact in emergency settings. Two complementary evaluations are underway: (a) Live simulations quasi-experiments at a medical simulation centre comparing scenarios with and without the AI-powered PFA system, followed by debriefings on ethics, usability, and cultural sensitivity (Ethical approval: #OT10042025A); and (b) a natural experiment assessing ASR symptoms before and after use of the PFA app during real-time emergencies (Ethical approval: #GT18122024A).

PFA delivers immediate support after traumatic events, yet models differ in approach and principles (e.g. the SIX Cs, Farchi et al.; the RAPID model, Everly & Lating, 2022). Evidence on utilization varies (e.g. Levy et al., 2020), and the scientific support for impact on trauma survivors is mixed (e.g. Kaur et al., 2022; Maldonato et al., 2020). Although a diverse array of PFA interventions exists, each is backed by differing levels of empirical evidence, and no single model has yet garnered sufficient validation for broad implementation. The AI tool described here can assist PFA researchers in testing an integrated PFA. Accordingly, upcoming studies will prioritize user experience and effectiveness metrics, with follow-up feedback from first responders and trauma survivors to illuminate real-world impact and advance both PFA and e-health, particularly AI in medicine. Structured user studies will systematically gather field feedback and iteratively refine the tool based on real-world usage.

9. Limitations

Despite its potential to enhance the delivery of evidence-based PFA, the AI recommendation system has several important limitations. First, due to ethical and confidentiality constraints, the system does not monitor or track the trauma survivor's symptoms over time, nor does it adapt its recommendations based on evolving clinical presentations. This design choice safeguards privacy but limits the system's capacity for dynamic, personalized refinement. Second, while the system is currently deployed via web application in three languages, this platform is not universally adopted across all international contexts, potentially restricting broader global implementation. To deal with this issue, we have now started the development of the next version of the system that integrates alternative communication platforms. In the future, we hope to develop a multilingual interface to ensure accessibility and cultural relevance. Third, although the system is programmed to respond in English, Hebrew, and Arabic, challenges remain, particularly with Arabic. In some cases, refinements to both the LLM and the supporting recommendation tables are necessary to ensure culturally sensitive responses (Chow & Li, 2024). This is especially critical in the system's initial recommendation phase, where culturally appropriate expressions of commitment from the first responder to the survivor are vital. For instance, in Arabic-speaking contexts, a first responder may refer to an elderly male survivor as 'Ya Ami' (Oh my uncle) to convey respect and emotional closeness, as an important cultural nuance that may not yet be fully captured by the system's current configurations.

Disclosure statement

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Data availability statement

A pilot instance of the AI-PFA application is available (www.pfabot.com), enabling readers to experience the tool's interface and core functions. The dataset underpinning this study comprises 20 meticulously curated, real-world generic scenarios, developed and validated by trauma and emergency response experts to refine the AI-based PFA chatbot recommendations. The dataset is designed solely for research and development purposes and adheres to established PFA intervention protocols. Importantly, the data does not include any personal or identifiable information. The dataset is accessible via the following hyperlink (https://github.com/sdannyvi/PTSDBot/blob/main/cases_dataset%20translated%20to%20Eng.xlsx). Researchers wishing to utilize the dataset are required to properly cite this article in their work. For further information or

inquiries regarding data access, please contact the corresponding author.

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Appendix

1. User: It's Shalom, I rushed to a scene where two buildings collapsed during the earthquake. A woman was shouting for help, we got her out, and she's okay with no injuries, but she still can't speak and can't stop shaking PFA chatbot Hi Shalom, here are several suggestions on how to deal with the situation you presented.

   First: You need to frame the event for the woman who is in distress. Explain to her that the event has ended.

   Second: Make eye contact with her and tell her that you are with her now, that you won't leave until she's able to function again, that together you'll overcome the problem, and that you need her cooperation.

   Third: Ask her four questions that will help her engage her thinking but avoid emotion. For example: What's her full name? When did the earthquake happen? What was she doing before the earthquake? Who else was around when the earthquake happened? How many people were injured in the earthquake? After she answers the questions, give her positive feedback and tell her that her actions during the event were important.

   Fourth: Now you need to give her four simple tasks to help her regain functionality and give her positive feedback for each action. For example: Ask her to say her name out loud, to perform a physical action like standing up, walking, or pointing, to say the names of people who were involved in the event out loud, to take out her mobile phone or decide who she wants to call to update them on her location, where she is going and that she is receiving help.

   Explain to her that she needs to stay active and report who will be with her in the coming days and who she can call if she has difficulty performing tasks.

   Finally, Shalom, tell her an encouraging sentence, for example: "Your strength during the event is

helping those around you to recover from the situation." Good luck!

2. User: A child witnessed his mother being stabbed by his father. He is in complete shock and shivering. How can we help him? PFA chatbot: Hi Talia, here are a few recommendations on how to handle the situation you presented. PFA chatbot: First, you need to frame the event for the child who is in shock and explain that the event is over.

   PFA chatbot: Second, Talia, you should make eye contact with the child and reassure him that you are there with him now, that you won't leave him until he recovers, that together you will overcome the problem and that you need his cooperation. PFA chatbot: Third, ask the child four questions to stimulate his thinking but avoid evoking emotions. Examples could be: What's the name of your mom and dad? When did you last see your mom? When were you told about the incident? How long have you known your parents? After he answers the questions, give him positive feedback and tell him that his actions were important.

   PFA chatbot: Now, Talia, have the child do three simple tasks that will get him back to function and routine. Make sure you give him positive feedback on each action. Examples could be: Find your phone and call someone you would like to update about the situation, choose who you would like to accompany you and build a schedule for the upcoming days together, update and choose together who will emotionally care for you in the days following the event. PFA chatbot: Talia, remind the child to stay active and maintain a routine as much as possible, find out who will be with him in the upcoming days and who he can call in case he struggles to perform tasks or go to school. PFA chatbot: Finally, Talia, tell him something encouraging, for example: Your resilience during this event is helping those around you to get out of the situation. Good luck!
3. Framing the event for a child should be age-appropriate. *For example: Ages 4–6*: 'You were at home. A very bad thing happened, and your parent was hurt and died. You are safe with us. Next, we're going to a quiet room, and Aunt Rana will meet us there.' *Ages 10–12*: 'There was a violent event at home, and your parent was killed. You're safe here. Next, we'll move to a private room; a staff member you know will come, and we'll call your other caregiver. I'll write the steps so you can see the plan.'
4. User: I arrived at the scene. Two kids, around 8 and 9 years old. The ambulance driver reported that the kids saw a man fall from a crane near the school. The kids are shaking. They are completely in shock and unable to speak.

   PFA chatbot: What is your first name, please?

   User: Danny PFA chatbot: I will address your request immediately.

   PFA chatbot: Hi Danny, here are some recommendations on how to deal with the situation you presented.

   PFA chatbot: First step: Danny, you need to frame the event for the children. Tell them that the event where the man fell from the crane is over.

   PFA chatbot: Second step: Danny, you need to make eye contact with the children and tell them that you are with them now, that you won't leave them until they return to functioning, that together you will overcome the problem, and that you need their cooperation.

   PFA chatbot: Third step: Danny, ask the children four questions that will help them engage their thinking. Examples: When did the event happen? What were they doing before the man fell from the crane? Who else was near them when the event happened? What colour was the shirt of the man who fell? After the children answer the questions, give them positive feedback and tell them that their actions during the event were important.

   PFA chatbot: Fourth step: Danny, give the children four simple tasks to get them back into action and routine, and give them positive feedback for each action. Examples: Have them say their names out loud, stand up and point to where the man fell, check where their belongings are, count how many things they have in their bags, decide which parent to inform about what happened. Encourage the children to return to activity. Danny, report to the children's parents and decide together on further support and professional assistance in the coming days to help the children return to normalcy. PFA chatbot: Thank you for contacting me. For further assistance, you can contact ERAN at 2201*.